another object, the electron and screen are now together in a superposition of configurations, and it is only at the moment of further interaction with a third object that their shared cloud of probability "collapses" and materializes in a particular configuration—and so on.

It is hard to take in the idea that an electron behaves in such a bizarre way. It is even more difficult to digest that this is also the way time and space behave. And yet, according to all the evidence, this is the way the quantum world works: the world that we inhabit.

The physical substratum that determines duration and physical intervals—the gravitational field—does not only have a dynamic influenced by masses; it is also a quantum entity that does not have determined values until it interacts with something else. When it does, the durations are granular and determinate only for that something with which it interacts; they remain indeterminate for the rest of the universe.

Time has loosened into a network of relations that no longer holds together as a coherent canvas. The picture of spacetimes (in the plural) fluctuating, superimposed one above the other, materializing at certain times with respect to particular objects, provides us with a very vague vision. But it is the best that we have for the fine granularity of the world. We are peering into the world of quantum gravity.

Let me reprise the long dive into the depths made in the first part of this book. There is no single time: there is a different duration for every trajectory; and time passes at different rhythms according to place and according to speed. It is not directional: the difference between past and future does not exist in the elementary equations of the world; its orientation is merely a contingent aspect that appears when we look at things and neglect the details. In this blurred view, the past of the universe was in a curiously "particular" state. The notion of the "present" does not work: in the vast universe there is nothing that we can reasonably call "present." The substratum that determines the duration of time is not an independent entity, different from the others that make up the world; it is an aspect of a dynamic field. It jumps, fluctuates, materializes only by interacting, and is not to be found beneath a minimum scale. . . . So, after all this, what is left of time?

> You got to deep-six your wristwatch, you got to try and understand,
> The time it seems to capture is just the movement of its hands . . . [57]